Video Transcript - Part 12

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Hey everyone, welcome to part twelve. In the last video we got an overview of SageMaker. Now we're going to walk through the core of the workflow in more detail: processing your data, training a model, and deploying it, all with SageMaker AI. If you understand these three steps, you understand how most machine learning projects run on AWS.

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Here's the plan. We'll start with the big picture of how data flows through these stages. Then we'll cover data processing, including which tool to pick, followed by training and the options you have there. After that, we'll look at deployment, and finish by seeing how these steps can be automated with pipelines.

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Let's start with the big picture. Almost everything in SageMaker starts and ends with Amazon S3. Your raw data sits in S3. A processing job cleans it and writes the result back to S3. A training job reads that processed data and writes a trained model to S3. Then you deploy that model to an endpoint, which applications call to get predictions. S3 is the hub in the middle, and each stage hands its output to the next.

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First, data processing. A SageMaker Processing job is a managed way to run your data preparation code. You give it an input location in S3, a script, and a container, and SageMaker launches the compute, runs your script, writes the output back to S3, and shuts everything down. You pay only while it runs. There are ready made containers for scikit-learn and Spark, or you can bring your own. People use processing jobs for cleaning data, feature engineering, splitting data into training and test sets, and even evaluating a model after training.

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SageMaker isn't the only option for preparing data, so it helps to know when to use what. A processing job is great when you have a script you want to run on demand or on a schedule. Data Wrangler is the visual option, where you point and click to transform data with little or no code. AWS Glue is a serverless ETL service, a good fit for moving and transforming data across many sources. And Amazon EMR with Spark is the choice for very large scale data processing. All four can read from and write to S3, so they fit together nicely.

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Next is training. A training job follows four steps. First, you point to your training data in S3. Second, you choose your algorithm, which is delivered as a container image, and pick the instance type and count. Third, SageMaker launches the instances, brings in the data, and trains

the model. Fourth, it saves the finished model artifact, a file called model dot tar dot gz, back to S3, and shuts the instances down. Logs and metrics go to CloudWatch, so you can see how training went.

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You have a lot of flexibility in how you train. For algorithms, you can use a built in one like XGBoost, run your own script with a framework like PyTorch or TensorFlow, or bring your own container. For data input, file mode copies the whole dataset to the training instance first. FastFile mode streams files from S3 as they're needed, so training can start sooner. And pipe mode streams data directly into the algorithm, which works well for very large datasets. For big models or data, distributed training spreads the work across many instances, and managed spot training plus automatic model tuning help you cut cost and improve accuracy.

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Now deployment. Deploying a model in SageMaker involves three pieces. First, the model itself, which combines your model artifact from S3 with an inference container that knows how to run it. Second, an endpoint configuration, where you choose the instance type, the instance count, and any variants. And third, the endpoint, which is the live HTTPS address your applications call to get predictions. You create them in that order, and SageMaker handles launching and managing the servers behind it.

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Once you're live, SageMaker gives you tools to run endpoints safely and cheaply. Auto scaling adds or removes instances as traffic goes up and down. Production variants let you send a share of traffic to a new model version, so you can run an A/B test. Multi model endpoints host many models behind one endpoint, which saves money when you have lots of models that are each used rarely. And deployment guardrails like blue green and canary rollouts let you shift traffic gradually, and roll back if something goes wrong. As we covered last time, you can also choose serverless, asynchronous, or batch inference when a real time endpoint isn't the right fit.

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These steps don't have to be manual. SageMaker Pipelines lets you chain them into one repeatable workflow. A typical pipeline processes the data, trains the model, evaluates it, and only if the results meet your bar, registers the model in the Model Registry and deploys it. Every run is tracked, so you can reproduce results, and you can trigger the pipeline on a schedule or when new data arrives.

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So to recap, data processing, training, and deployment are the heart of the SageMaker workflow, and S3 connects them. Processing jobs prepare your data, training jobs turn it into a model artifact, and endpoints serve that model to your applications. Pipelines tie it all together so the process is repeatable. Thanks for watching, and I'll see you in the next one.